JK Lakshmipat University Jaipur
INSTITUTE OF ENGINEERING AND TECHNOLOGY

Mid Term Examination, September 2026

B. Tech. Semester I (Section D and H)

Roll No. 519

Course Code: EE1118 Fundamental of Electrical and Electronic Engineering **Time: 2:00 hours** **Max. Marks: 15**

Instructions to students:

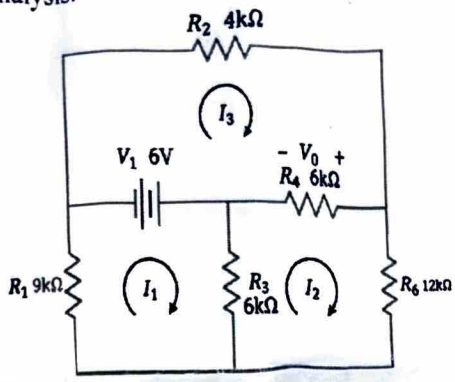
1. Do not write anything other than your roll number on question paper.
2. Assume suitable data wherever required and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics/ flow chart wherever, required are an integral part of the answer.

Answer all questions.

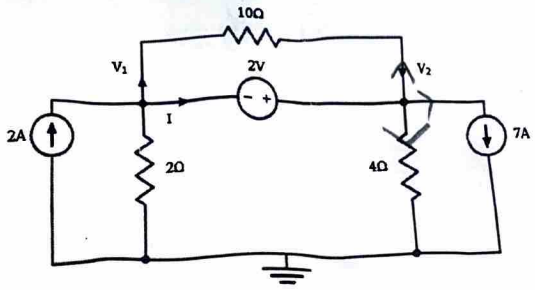
Each question carries equal marks.

***Nonprogrammable calculator allowed**

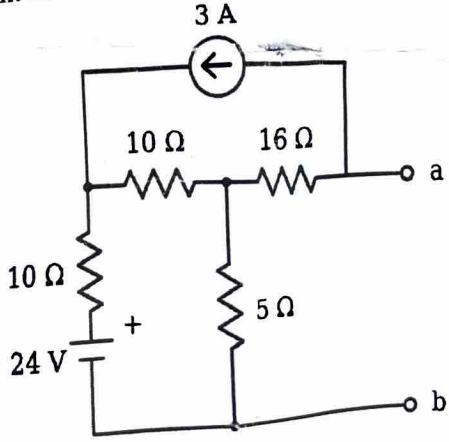
1. Find V_o using mesh analysis. *(03) CO1 (BTL 2 & 3)*



2. Using nodal analysis find the node voltages V_1 and V_2. *(03) CO1 (BTL 2 & 3)*

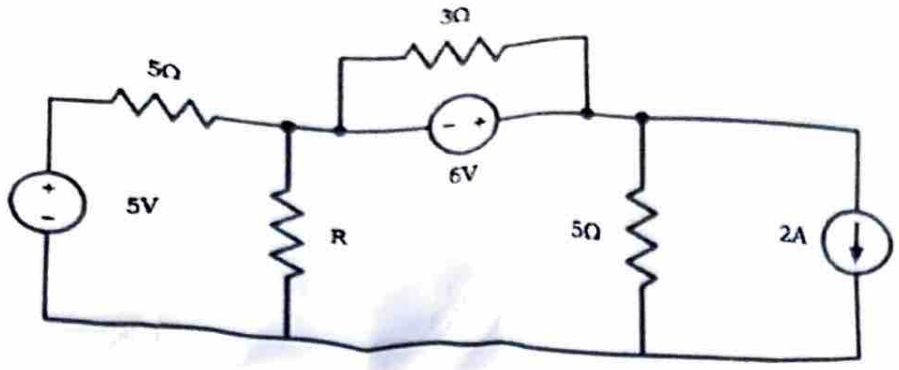


3. Draw Thevenin equivalent network of the given circuit across terminal a-b. *(03) CO1 (BTL 3,5 and 6)*



4. What is the maximum power transferred to R?

(03)
CO1
(BTL 3 & 5)



5. The switch in the circuit shown below has been closed for long time. At t=0, the switch is opened. Find the voltage across the capacitor at $t = \tau,\ 2\tau$.

(03)
CO2
(BTL 4 & 5)

